

05 Inverse of a Matrix

5.1 The Inverse Matrix

Understanding the matrix inverse through geometry: undoing scaling, rotation, shear, and reflection



Linear Algebra Made Easy and Visual | Open-source code & drafts: <https://github.com/Visualize-ML>

Core Skills You Will Master

1

Invertible matrix: a square matrix with nonzero determinant

3

A square matrix is not invertible when the transformation collapses dimensions or loses information

5

The inverse of a rotation is a rotation by the opposite angle, and rotation matrices are orthogonal

7

A reflection is its own inverse

2

A matrix is a linear transformation; its inverse is that transformation's "undo" operation, restoring the original geometric structure — the determinant reflects the change in area

4

The inverse of a scaling matrix restores the original position through reciprocal scaling

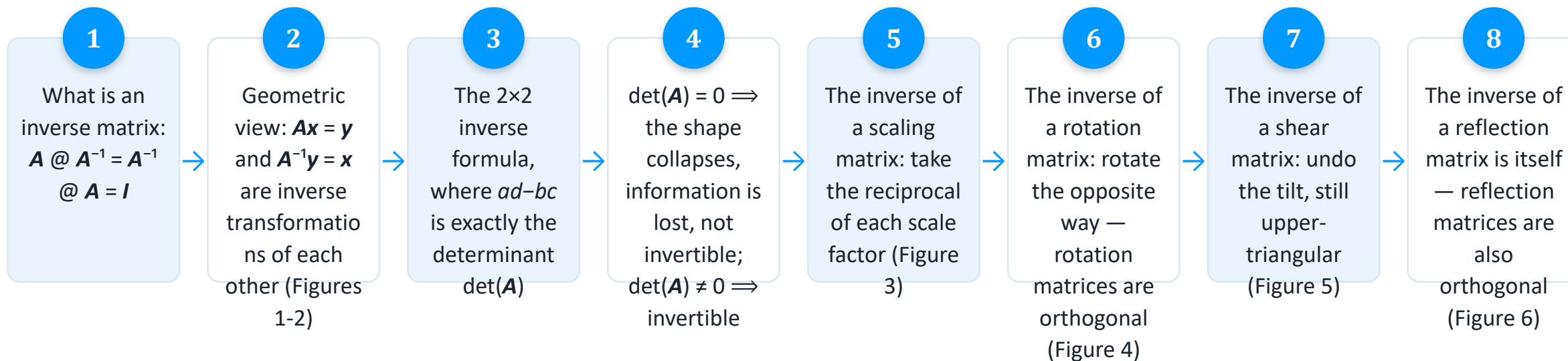
6

The inverse of a shear undoes the tilt to restore the original shape

8

Orthogonal matrices include rotations, reflections, permutations, and the identity matrix

How This Section Unfolds



What Is an Inverse Matrix

$$\mathbf{A} @ \mathbf{A}^{-1} = \mathbf{A}^{-1} @ \mathbf{A} = \mathbf{I}$$

- For $\mathbf{A}$ to have an inverse, $\mathbf{A}$ must be square — equal rows and columns; but not every square matrix has an inverse
- For an $n \times n$ matrix $\mathbf{A}$, if a matrix $\mathbf{A}^{-1}$ exists satisfying the equation above ($\mathbf{I}$ is the $n \times n$ identity matrix), $\mathbf{A}$ is called an invertible matrix, or a non-singular matrix
- Computing an inverse by hand is arguably the least illuminating part of the topic — the formula itself doesn't explain what the inverse really means, so we reach for the geometric view instead

Figure 1 · Column Vector $\mathbf{x}$ Maps to Column Vector $\mathbf{y}$ Under a 2×2 Matrix $\mathbf{A}$

- For an $n \times n$ matrix $\mathbf{A}$, $\mathbf{Ax} = \mathbf{y}$ can be seen as a geometric transformation — scaling, rotation, shear, reflection, or some sequence of these
- The column vector $\mathbf{x}$ is the Input; after mapping through $\mathbf{A}$, we get the Output column vector $\mathbf{y}$

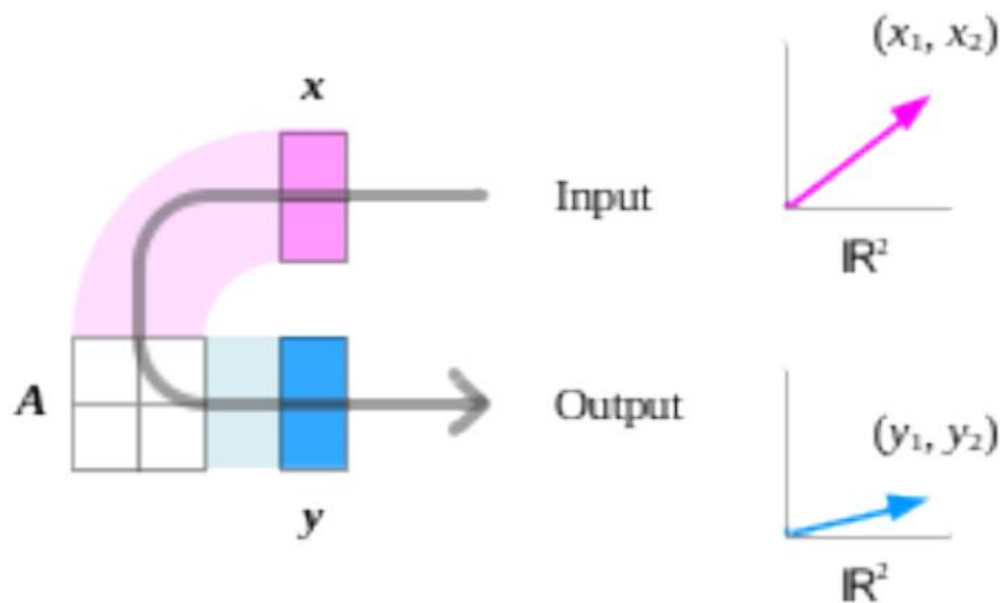


图 1. 列向量 $\mathbf{x}$ 在 2×2 方阵 $\mathbf{A}$ 映射下结果为列向量 $\mathbf{y}$

Figure 2 · Column Vector y Maps to Column Vector x Under a 2×2 Matrix A Inverse

- $A^{-1}y = x$ can be seen as the inverse of A 's transformation — mapping the output y back to the input x
- This only works when A is invertible; otherwise, A 's geometric transformation cannot be undone

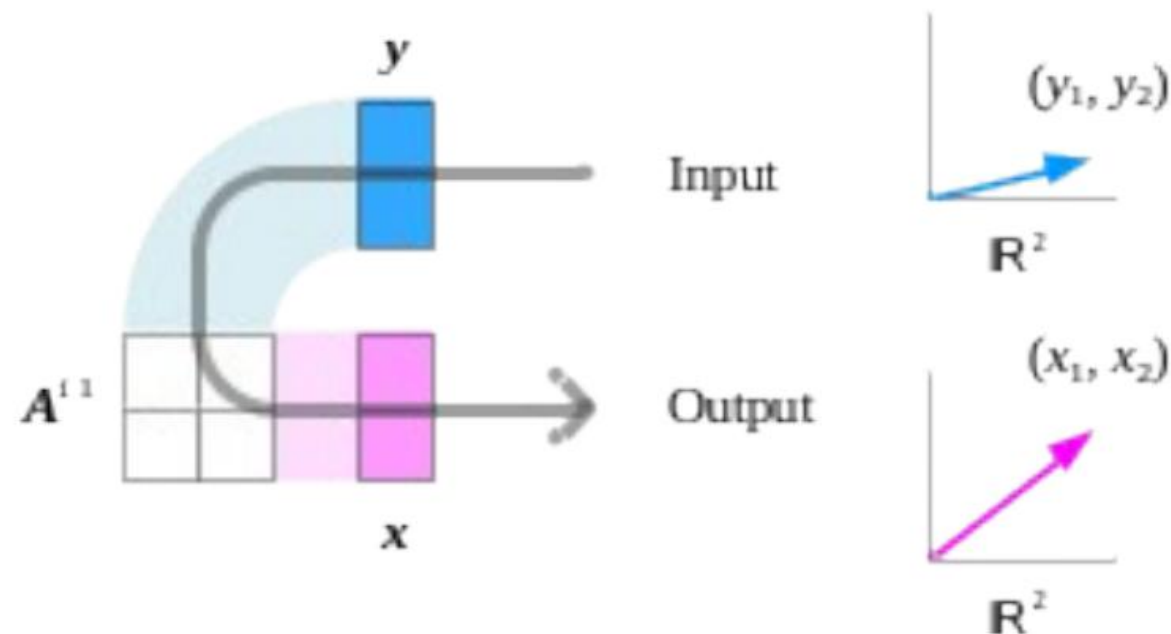


图 2. 列向量 y 在 2×2 方阵 A^{-1} 映射下结果为列向量 x

Take a 2×2 Matrix as an Example

$$\mathbf{A} = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \Rightarrow \mathbf{A}^{-1} = \begin{bmatrix} d & -b \\ -c & a \end{bmatrix} / (ad-bc)$$

- As long as $ad-bc \neq 0$, the inverse exists — and $ad-bc$ is exactly the determinant $\det(\mathbf{A})$!
- Earlier in this book we noted that the determinant represents area or volume; if $\det(\mathbf{A}) = 0$, the linear transformation collapses the shape, some information is lost, and the matrix has no inverse
- If $\det(\mathbf{A}) \neq 0$, the shape doesn't collapse, no information is lost, and the transformation is invertible

CODE 1

A Custom Function to Invert a 2×2 Matrix

LA_Ch05_01_01.ipynb

- a) Define `inverse_2x2_A(A)`, which computes the inverse of a 2×2 matrix and checks whether it's invertible
- b) Extract `a, b` from the first row of `A`, and `c, d` from the second row
- c) Compute the determinant `det = a*d - b*c`, used to check invertibility
- d) If `det` equals 0, print a message and return `None` — the matrix has no inverse
- e) Build a 2×2 array with entries `d, -b, -c, a`, then divide by `det` to get `A`'s inverse
- f) Create a matrix `A`, then call the custom function to compute its inverse

```
## 初始化
import numpy as np

## 自定义函数

def inverse_2x2_A(A):
    """
    A = [[a, b],
         [c, d]]
    """
    # 提取矩阵元素
    a, b = A[0]
    c, d = A[1]

    # 计算行列式 det(A) = ad - bc
    det = a * d - b * c

    # 判断行列式是否为零
    if det == 0:
        print("矩阵不可逆, 行列式为零")
        return None

    # 计算逆矩阵
    inv_A = np.array([[d, -b], [-c, a]]) / det
    return inv_A

## 定义矩阵
A = [[1, 2],
     [3, 4]]
A = np.array(A)

## 计算逆矩阵
inverse_2x2_A(A)
```

代码 1. 2×2 矩阵的逆 | LA_Ch05_01_01.ipynb

The Inverse of a Scaling Matrix Scales by the Reciprocal Ratio

$$\mathbf{S} = \begin{bmatrix} 3/2 & 0 \\ 0 & 2/3 \end{bmatrix} \Rightarrow \mathbf{S}^{-1} = \begin{bmatrix} 2/3 & 0 \\ 0 & 3/2 \end{bmatrix}$$

- $\mathbf{S}$ stretches the x_1 coordinate by 1.5× while shrinking the x_2 coordinate to 2/3; but $\det(\mathbf{S}) = 1$, so this scaling doesn't change the area
- Each scale factor in $\mathbf{S}^{-1}$ is the reciprocal of the original: x_1 shrinks to 2/3 while x_2 stretches by 1.5×, exactly undoing the original transformation

Figure 3 · Scaling and Its Inverse in the Plane, a 2×2 Matrix

- The square is stretched and compressed into a new shape by S ; applying S^{-1} fully restores the original square
- Any diagonal matrix is invertible as long as none of its diagonal entries is zero — its inverse is also diagonal, with reciprocal entries

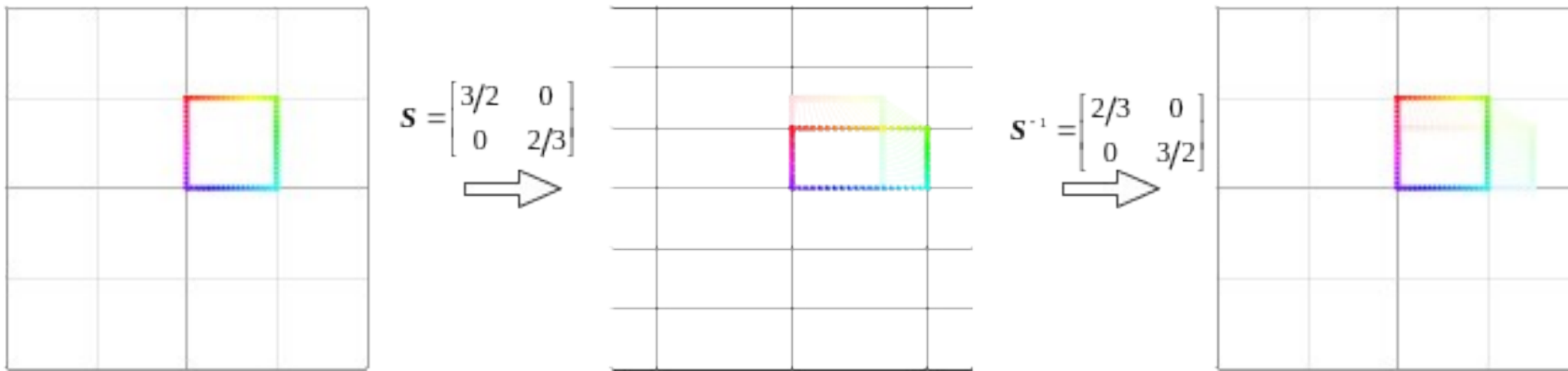


图 3. 平面上的缩放、缩放的逆操作， 2×2 矩阵

$$\mathbf{S} @ \mathbf{S} \text{ Inverse} = \mathbf{S} \text{ Inverse} @ \mathbf{S} = \mathbf{I}$$

$$\mathbf{S} @ \mathbf{S}^{-1} = \mathbf{S}^{-1} @ \mathbf{S} = \mathbf{I}$$

- Multiplying the two gives the identity matrix — no change at all; scaling followed by its inverse brings the shape back to where it started
- A matrix's inverse often shares its structure: here, every nonzero entry on the main diagonal is simply the reciprocal of the original entry

The Inverse of a Rotation Is a Rotation the Other Way

$$\mathbf{R} = \begin{bmatrix} -1/2 & -\sqrt{3}/2 \\ \sqrt{3}/2 & -1/2 \end{bmatrix} \Rightarrow \det(\mathbf{R}) = 1, \mathbf{R}^{-1} = \begin{bmatrix} -1/2 & \sqrt{3}/2 \\ -\sqrt{3}/2 & -1/2 \end{bmatrix}$$

- $\mathbf{R}$ rotates every vector in the plane 120° counterclockwise about the origin, leaving the origin fixed; its determinant is 1, so the transformation doesn't change area
- $\mathbf{R}$ represents a 120° counterclockwise rotation, so its inverse $\mathbf{R}^{-1}$ has the same structure but the opposite angle — a 120° clockwise rotation

Figure 4 · Rotation and Its Inverse in the Plane, a 2×2 Matrix

- The square is rotated 120° counterclockwise by R ; applying R^{-1} rotates it 120° clockwise, returning it to its original position
- If a point is rotated 120° counterclockwise by R and then 120° clockwise by R^{-1} , it lands right back where it started

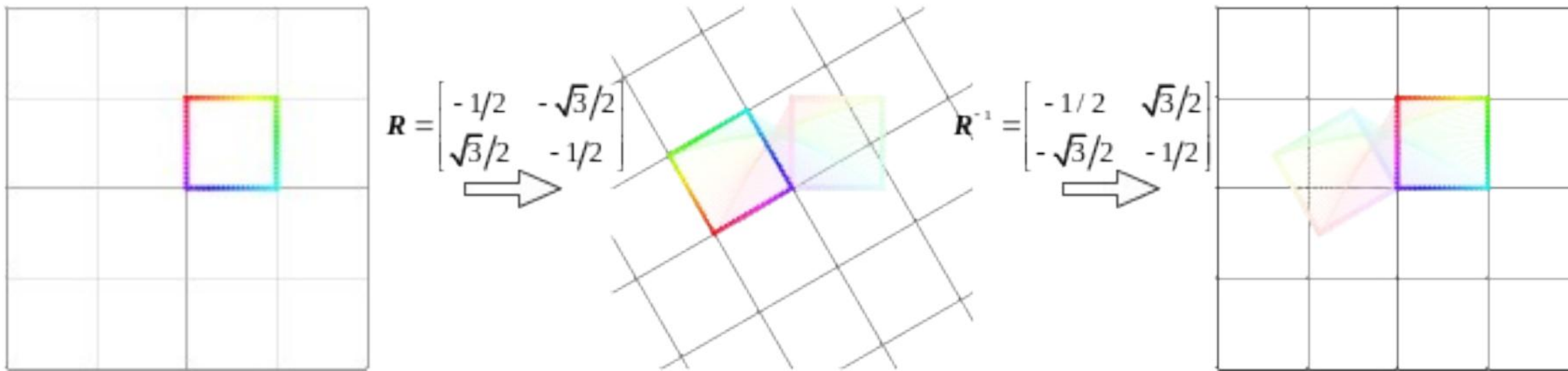


图 4. 平面上的旋转、旋转的逆操作， 2×2 矩阵

The Rotation Matrix's Transpose Equals Its Inverse

$$\mathbf{R} @ \mathbf{R}^{-1} = \mathbf{I}, \mathbf{R}^T = \mathbf{R}^{-1}$$

- There's a relationship worth noting for rotation matrices: $\mathbf{R}^T = \mathbf{R}^{-1}$, which means $\mathbf{R}^T \mathbf{R} = \mathbf{I}$ and $\mathbf{R} \mathbf{R}^T = \mathbf{I}$
- Any square matrix satisfying this relationship is called an orthogonal matrix — this book will cover orthogonal matrices in more depth later on

The Inverse of a Shear Undoes the Tilt

$$\mathbf{K} = \begin{bmatrix} 1 & 0.5 \\ 0 & 1 \end{bmatrix} \Rightarrow \det(\mathbf{K}) = 1, \mathbf{K}^{-1} = \begin{bmatrix} 1 & -0.5 \\ 0 & 1 \end{bmatrix}$$

- A shear matrix tilts the coordinate system along a given direction, so one component of a vector is stretched or compressed by the other, without changing its length
- $\mathbf{K}$ shears along x_1 : the x_2 component is unchanged, while the x_1 component gains 0.5 times the x_2 coordinate; $\mathbf{K}^{-1}$ subtracts that same amount, undoing the shear

Figure 5 · Shear and Its Inverse in the Plane, a 2×2 Matrix

- The square is tilted into a parallelogram by K ; applying K^{-1} undoes the tilt, restoring the original square
- Notice that both K and K^{-1} are upper-triangular — the inverse of an upper-triangular matrix is still upper-triangular

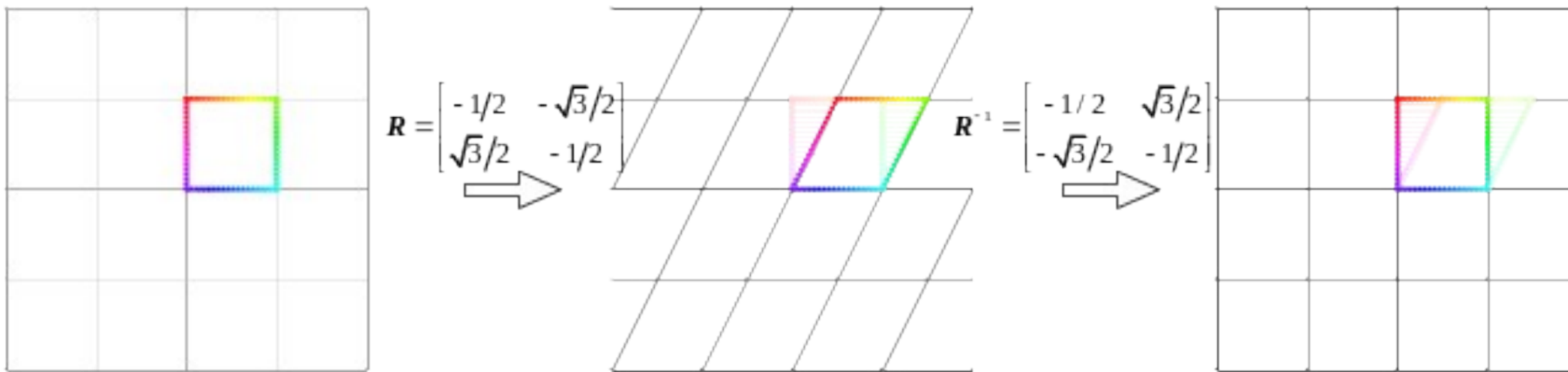


图 5. 平面上的剪切、剪切的逆操作， 2×2 矩阵

A Reflection Matrix Is Its Own Inverse

$$\mathbf{M} = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} \Rightarrow \det(\mathbf{M}) = -1, \mathbf{M}^{-1} = \mathbf{M}$$

- $\mathbf{M}$ reflects every vector across the x_2 axis: the x_1 coordinate flips sign (a horizontal flip), while the x_2 coordinate stays the same
- The absolute value of the determinant is 1, so area is unchanged; but the determinant is negative, meaning orientation flips — counterclockwise, $\mathbf{e}_1$ leads $\mathbf{e}_2$ before the transform, but $\mathbf{e}_2$ leads $\mathbf{e}_1$ after $\mathbf{M}$ is applied

Figure 6 · Reflection and Its Inverse in the Plane, a 2×2 Matrix

- The square is flipped horizontally by $\mathbf{M}$; applying the same reflection again flips it right back to its original position
- This matches intuition: if a point is flipped once by a reflection, applying that same reflection again returns it to where it started

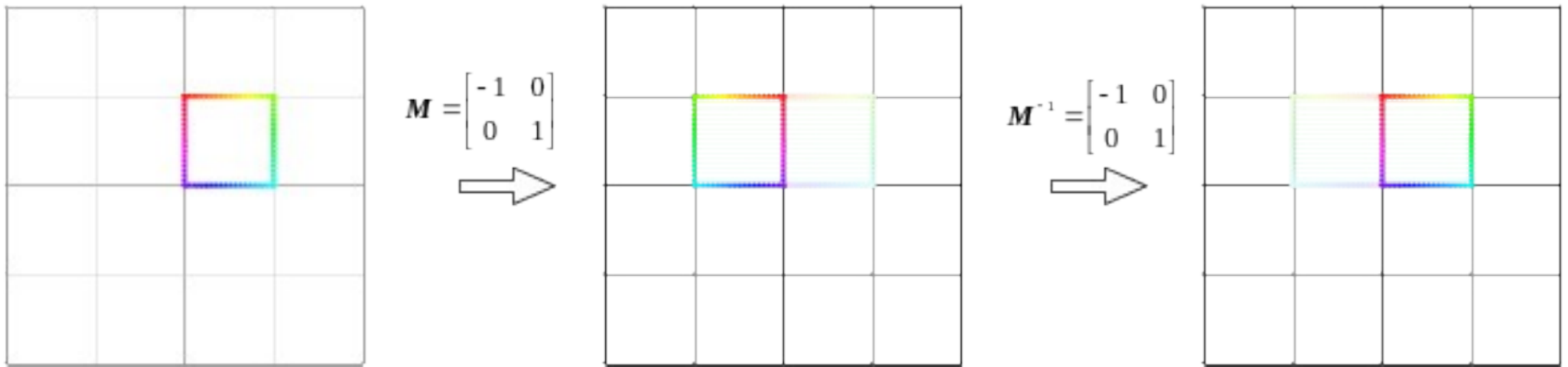


图 6. 平面上的镜像、镜像的逆操作， 2×2 矩阵

M's Transpose Also Equals Itself

$$\mathbf{M} @ \mathbf{M}^{-1} = \mathbf{I}, \mathbf{M} = \mathbf{M}^{-1}$$

- A reflection is symmetric: applying the same reflection twice restores the original position, so a reflection matrix is its own inverse
- Reflection matrices are also orthogonal: $\mathbf{M}^T \mathbf{M} = \mathbf{I}$ and $\mathbf{M} \mathbf{M}^T = \mathbf{I}$ — its transpose equals its inverse

Invertibility Depends on Whether Information Is Preserved

$\det(A) \neq 0 \Leftrightarrow A$ is invertible $\Leftrightarrow$ the transformation doesn't collapse or lose information

- Scaling's inverse is reciprocal scaling, rotation's inverse is rotating the other way, shear's inverse undoes the tilt, and reflection is its own inverse — each of the four basic transformations has its own geometric intuition
- Rotation and reflection matrices both satisfy $\mathbf{Q}^T \mathbf{Q} = \mathbf{I}$ and are orthogonal; permutation matrices and the identity matrix are orthogonal too, and this book covers orthogonal matrices further later on
- The matrix inverse, determinant, rank, and linear dependence are all closely related concepts — the next section continues exploring further properties of the inverse

Key Terms in This Section

Inverse Matrix A^{-1} — satisfies $A @ A^{-1} = A^{-1} @ A = I$

Singular Matrix A square matrix with determinant 0 — has no inverse

Orthogonal Matrix A square matrix satisfying $Q^T Q = Q Q^T = I$; represents a rotation or reflection

Inverse of a Rotation Matrix A rotation by the opposite angle, equal to the matrix's own transpose

Inverse of a Reflection Matrix A reflection matrix is involutory, satisfying $M = M^{-1}$

Invertible / Non-singular Matrix A square matrix with nonzero determinant

Identity Matrix I — satisfies $A @ I = I @ A = A$, the "no change" baseline in the inverse's definition

Inverse of a Scaling Matrix Take the reciprocal of each nonzero diagonal entry — still diagonal

Inverse of a Shear Matrix Undoes the tilt; K and K^{-1} are both upper-triangular

Involutory Matrix A square matrix satisfying $M^2 = I$ — a reflection matrix is a typical example

Code and Further Reading

Further Reading · `numpy.linalg.inv`

LA_05_01_02.ipynb shows how to compute a matrix inverse with `numpy.linalg.inv()` — self-study recommended

Further Reading · Symbolic Inversion with SymPy

LA_05_01_03.ipynb introduces computing a matrix inverse symbolically with SymPy — self-study recommended

Further Reading · Orthogonal Matrices

This book covers orthogonal matrices (rotation, reflection, permutation, identity) in more depth later on

Open-Source Repository

All code and figures for this book: <https://github.com/Visualize-ML>

Exercises Q1 – Q8

Q1 Determine whether each of the following matrices is invertible: ① $\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$; ② $\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$; ③ $\begin{bmatrix} 1 & 0 & 0 & 1 \\ 1 & 1 & 0 & 1 \end{bmatrix}$; ④ $\begin{bmatrix} 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 1 \end{bmatrix}$; ⑤ $\begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 0 & 0 & 1 \end{bmatrix}$.

Q3 Compute the inverse of each shear matrix below, and describe its geometric operation: ① $\begin{bmatrix} 1 & 1 & 0 & 1 \\ 1 & -1 & 0 & 1 \end{bmatrix}$; ② $\begin{bmatrix} 1 & 0 & 1 & 1 \\ 1 & 0 & -1 & 1 \end{bmatrix}$.

Q5 Compute the inverse of each reflection matrix below, and describe its geometric operation: ① $\begin{bmatrix} 1 & 0 & 0 & -1 \\ 0 & 1 & 1 & 0 \end{bmatrix}$; ② $\begin{bmatrix} 0 & 1 & 1 & 0 \\ 0 & -1 & -1 & 0 \end{bmatrix}$.

Q7 Use `numpy.linalg.inv()` to verify that the inverse of this section's scaling matrix $S = \begin{bmatrix} 3/2 & 0 & 0 & 2/3 \\ 2/3 & 0 & 0 & 3/2 \end{bmatrix}$ equals $S^{-1} = \begin{bmatrix} 2/3 & 0 & 0 & 3/2 \end{bmatrix}$.

Q2 Compute the inverse of each scaling matrix below, and describe its geometric operation: ① $\begin{bmatrix} 1 & 0 & 0 & 2 \\ 3 & 0 & 0 & 1 \end{bmatrix}$; ② $\begin{bmatrix} 3 & 0 & 0 & 1 \\ 3 & 0 & 0 & 2 \end{bmatrix}$.

Q4 Compute the inverse of each rotation matrix (about the origin) below, and describe its geometric operation: ① $\begin{bmatrix} \sqrt{3}/2 & -1/2 & 1/2 & \sqrt{3}/2 \\ \sqrt{2}/2 & -\sqrt{2}/2 & \sqrt{2}/2 & \sqrt{2}/2 \end{bmatrix}$; ② $\begin{bmatrix} 0 & -1 & 1 & 0 \\ -1 & 0 & 0 & -1 \end{bmatrix}$.

Q6 Write `is_invertible(A)` that checks whether a square matrix A is invertible by testing whether its determinant is close to 0 (tolerance $1e-9$).

Q8 Write `verify_inverse(A)` that computes $A @ A^{-1}$ and $A^{-1} @ A$, and checks whether both equal the identity matrix (tolerance $1e-9$).

Exercises Q9 – Q17

Q9 Construct a 3×3 diagonal matrix $D=\text{diag}(2,3,4)$, compute D^{-1} by hand, and verify it with `numpy.linalg.inv()`.

Q11 Use `numpy.linalg.inv()` to compute the inverse of `rotation_matrix(np.pi/3)`, and verify it equals the matrix's transpose (an orthogonal-matrix property).

Q13 Construct a singular matrix (determinant 0) A , try computing `numpy.linalg.inv(A)`, observe the error raised, and explain why.

Q15 The reflection matrix M satisfies $M=M^{-1}$ — such matrices are called involutory. Construct two more 2×2 matrices satisfying $M^2=I$ and verify with Python.

Q17 Use SymPy to invert a 2×2 symbolic matrix $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$, print the symbolic inverse expression, and compare it to this section's formula.

Q10 Write `rotation_matrix(theta)` that returns the 2×2 matrix for a counterclockwise rotation of θ radians about the origin, and verify its inverse equals `rotation_matrix(-theta)`.

Q12 Write `is_orthogonal(Q)` that checks whether a square matrix Q satisfies $Q^T Q = I$, to determine whether Q is orthogonal.

Q14 Write `shear_matrix(k)` that returns the 2×2 shear matrix with shear factor k along x_1 , and verify its inverse is `shear_matrix(-k)`.

Q16 Write `compose_transform(A, B, x)` that applies A then B to vector x , then applies $(AB)^{-1}$ to map the result back to x , and verify it recovers x .

Exercises Q18 – Q25

Q18 Write `random_invertible_matrix(n)` that randomly generates an $n \times n$ matrix, checking (and regenerating if needed) that its determinant is nonzero.

Q20 Write `visualize_inverse(A)` that uses matplotlib to plot the unit square transformed by A , and then by A^{-1} , verifying it returns to the original square.

Q22 Write `matrix_power_inverse(A, n)` that computes the inverse of A raised to the n -th power (A^n), and verify it equals $(A^{-1})^n$.

Q24 Write `condition_check(A)` that computes the condition number of A (`np.linalg.cond`), and discuss the numerical-stability issues that arise when it's very large.

Q19 For the upper-triangular matrix $K = \begin{bmatrix} 1 & 0.5 \\ 0 & 1 \end{bmatrix}$, verify that K^{-1} is still upper-triangular; do the same for a lower-triangular matrix.

Q21 Given the scaling matrix $S = \begin{bmatrix} 2 & 0 \\ 0 & 0.5 \end{bmatrix}$, compute how S and S^{-1} each transform the unit circle, and visually compare the two areas.

Q23 Construct 2×2 matrices A and B , and verify that $(AB)^{-1}$ equals $B^{-1}A^{-1}$ (not $A^{-1}B^{-1}$), explaining why with concrete numbers.

Q25 Drawing on this section, write a short summary (code comments are fine) of the geometric intuition behind the inverses of scaling, rotation, shear, and reflection.

10 Prompts to Explore Further (1/2)

1 Explain why only square matrices can have an inverse — why don't rectangular matrices have a (two-sided) inverse?

2 Explain the geometric meaning of a matrix inverse — why is $A^{-1}y=x$ the "undo" of $Ax=y$?

3 Explain why $\det(A)=0$ means A has no inverse — what does this have to do with "losing information"?

4 Explain why the inverse of a scaling matrix takes the reciprocal of each scale factor.

5 Explain why a rotation matrix's inverse equals its transpose, and how this connects to the definition of an orthogonal matrix.

10 Prompts to Explore Further (2/2)

6

Explain why the inverse of a shear matrix is still upper-triangular.

7

Explain why a reflection matrix is its own inverse, and what other matrices share this property ($M^2=I$).

8

Explain which geometric transformations orthogonal matrices include, and what they have in common.

9

Explain the logic behind Code 1's 2×2 inverse formula — why does it check the determinant first?

10

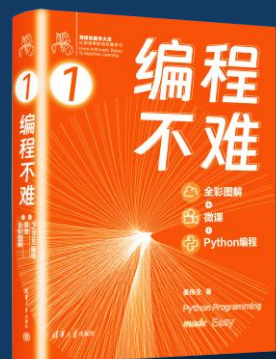
Drawing on this section, discuss the relationship between a matrix's invertibility and whether its linear transformation preserves all the information in space.

数学

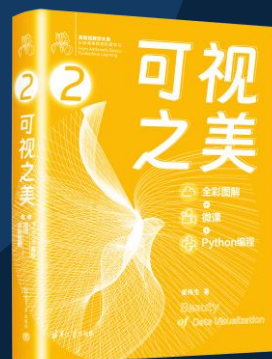
数学基础

线性代数

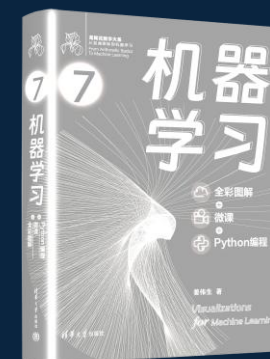
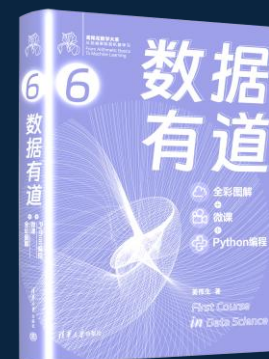
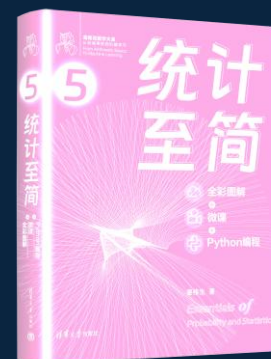
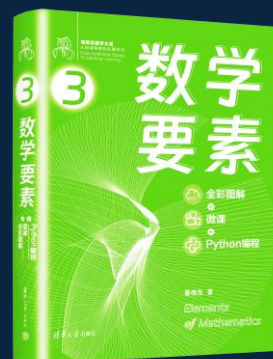
概率统计



Python编程



数据可视化



工具

数据分析
图和网络

回归、降维
分类、聚类

实践

